# Hierarchical_Clustering

May 30, 2026

[158]:
```
import pandas as pd
from sklearn.decomposition import PCA
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import AgglomerativeClustering
from scipy.cluster.hierarchy import linkage,dendrogram
import matplotlib.pyplot as plt
```

[159]:
```
df = pd.read_csv(
    "/home/umair/Downloads/OnlineRetail.csv",
    encoding='latin1'
)
df
```

[159]:

| | InvoiceNo | StockCode | Description | Quantity | \ |
|---|---|---|---|---|---|
| 0 | 536365 | 85123A | WHITE HANGING HEART T-LIGHT HOLDER | 6 | |
| 1 | 536365 | 71053 | WHITE METAL LANTERN | 6 | |
| 2 | 536365 | 84406B | CREAM CUPID HEARTS COAT HANGER | 8 | |
| 3 | 536365 | 84029G | KNITTED UNION FLAG HOT WATER BOTTLE | 6 | |
| 4 | 536365 | 84029E | RED WOOLLY HOTTIE WHITE HEART. | 6 | |
| … | … | … | … | … | |
| 541904 | 581587 | 22613 | PACK OF 20 SPACEBOY NAPKINS | 12 | |
| 541905 | 581587 | 22899 | CHILDREN'S APRON DOLLY GIRL | 6 | |
| 541906 | 581587 | 23254 | CHILDRENS CUTLERY DOLLY GIRL | 4 | |
| 541907 | 581587 | 23255 | CHILDRENS CUTLERY CIRCUS PARADE | 4 | |
| 541908 | 581587 | 22138 | BAKING SET 9 PIECE RETROSPOT | 3 | |

| | InvoiceDate | UnitPrice | CustomerID | Country |
|---|---|---|---|---|
| 0 | 12/1/2010 8:26 | 2.55 | 17850.0 | United Kingdom |
| 1 | 12/1/2010 8:26 | 3.39 | 17850.0 | United Kingdom |
| 2 | 12/1/2010 8:26 | 2.75 | 17850.0 | United Kingdom |
| 3 | 12/1/2010 8:26 | 3.39 | 17850.0 | United Kingdom |
| 4 | 12/1/2010 8:26 | 3.39 | 17850.0 | United Kingdom |
| … | … | … | … | … |
| 541904 | 12/9/2011 12:50 | 0.85 | 12680.0 | France |
| 541905 | 12/9/2011 12:50 | 2.10 | 12680.0 | France |
| 541906 | 12/9/2011 12:50 | 4.15 | 12680.0 | France |
| 541907 | 12/9/2011 12:50 | 4.15 | 12680.0 | France |
| 541908 | 12/9/2011 12:50 | 4.95 | 12680.0 | France |

```
[541909 rows x 8 columns]
```

[160]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 541909 entries, 0 to 541908
Data columns (total 8 columns):
 #   Column       Non-Null Count   Dtype
---  ------       --------------   -----
 0   InvoiceNo    541909 non-null  str
 1   StockCode    541909 non-null  str
 2   Description  540455 non-null  str
 3   Quantity     541909 non-null  int64
 4   InvoiceDate  541909 non-null  str
 5   UnitPrice    541909 non-null  float64
 6   CustomerID   406829 non-null  float64
 7   Country      541909 non-null  str
dtypes: float64(2), int64(1), str(5)
memory usage: 33.1 MB
```

[161]:
```
df.isnull().sum()
```

[161]:
```
InvoiceNo          0
StockCode          0
Description     1454
Quantity           0
InvoiceDate        0
UnitPrice          0
CustomerID     135080
Country            0
dtype: int64
```

[162]:
```
df.
  ↪drop(columns=["StockCode","Description","InvoiceDate","Country"],inplace=True)
df.dropna(subset=["CustomerID"], inplace=True)
df
```

[162]:

| | InvoiceNo | Quantity | UnitPrice | CustomerID |
|---|---|---|---|---|
| 0 | 536365 | 6 | 2.55 | 17850.0 |
| 1 | 536365 | 6 | 3.39 | 17850.0 |
| 2 | 536365 | 8 | 2.75 | 17850.0 |
| 3 | 536365 | 6 | 3.39 | 17850.0 |
| 4 | 536365 | 6 | 3.39 | 17850.0 |
| … | … | … | … | … |
| 541904 | 581587 | 12 | 0.85 | 12680.0 |
| 541905 | 581587 | 6 | 2.10 | 12680.0 |
| 541906 | 581587 | 4 | 4.15 | 12680.0 |

```
541907    581587         4       4.15     12680.0
541908    581587         3       4.95     12680.0

[406829 rows x 4 columns]
```

[163]:
```
df["TotalPrice"] = df["Quantity"]*df["UnitPrice"]
df
```

[163]:

| | InvoiceNo | Quantity | UnitPrice | CustomerID | TotalPrice |
|---|---|---|---|---|---|
| 0 | 536365 | 6 | 2.55 | 17850.0 | 15.30 |
| 1 | 536365 | 6 | 3.39 | 17850.0 | 20.34 |
| 2 | 536365 | 8 | 2.75 | 17850.0 | 22.00 |
| 3 | 536365 | 6 | 3.39 | 17850.0 | 20.34 |
| 4 | 536365 | 6 | 3.39 | 17850.0 | 20.34 |
| … | … | … | … | … | … |
| 541904 | 581587 | 12 | 0.85 | 12680.0 | 10.20 |
| 541905 | 581587 | 6 | 2.10 | 12680.0 | 12.60 |
| 541906 | 581587 | 4 | 4.15 | 12680.0 | 16.60 |
| 541907 | 581587 | 4 | 4.15 | 12680.0 | 16.60 |
| 541908 | 581587 | 3 | 4.95 | 12680.0 | 14.85 |

[406829 rows x 5 columns]

[164]:
```
customer_data = df.groupby("CustomerID").agg({
    "Quantity" : "sum" ,
    "TotalPrice": "sum",
    "InvoiceNo": "nunique"
})
customer_data.columns = [
    "TotalQuantity",
    "TotalSpent",
     "# of Invoices"
]
customer_data
```

[164]:

| CustomerID | TotalQuantity | TotalSpent | # of Invoices |
|---|---|---|---|
| 12346.0 | 0 | 0.00 | 2 |
| 12347.0 | 2458 | 4310.00 | 7 |
| 12348.0 | 2341 | 1797.24 | 4 |
| 12349.0 | 631 | 1757.55 | 1 |
| 12350.0 | 197 | 334.40 | 1 |
| … | … | … | … |
| 18280.0 | 45 | 180.60 | 1 |
| 18281.0 | 54 | 80.82 | 1 |
| 18282.0 | 98 | 176.60 | 3 |
| 18283.0 | 1397 | 2094.88 | 16 |

```
18287.0              1586     1837.28              3

[4372 rows x 3 columns]
```

[165]:
```
scaler = StandardScaler()
scaled_data = scaler.fit_transform(customer_data)
scaled_df = pd.DataFrame(scaled_data)
scaled_df
```

[165]:
```
              0         1         2
0    -0.240215 -0.231001 -0.329362
1     0.285870  0.293432  0.206102
2     0.260828 -0.012316 -0.115176
3    -0.105162 -0.017146 -0.436455
4    -0.198051 -0.190312 -0.436455
…         …         …         …
4367 -0.230583 -0.209026 -0.436455
4368 -0.228657 -0.221167 -0.436455
4369 -0.219240 -0.209513 -0.222269
4370  0.058784  0.023900  1.169939
4371  0.099236 -0.007444 -0.222269

[4372 rows x 3 columns]
```

[166]:
```
pca = PCA(n_components=2)
final_data = pca.fit_transform(scaled_df)
final_df = pd.DataFrame(final_data)
final_df
```

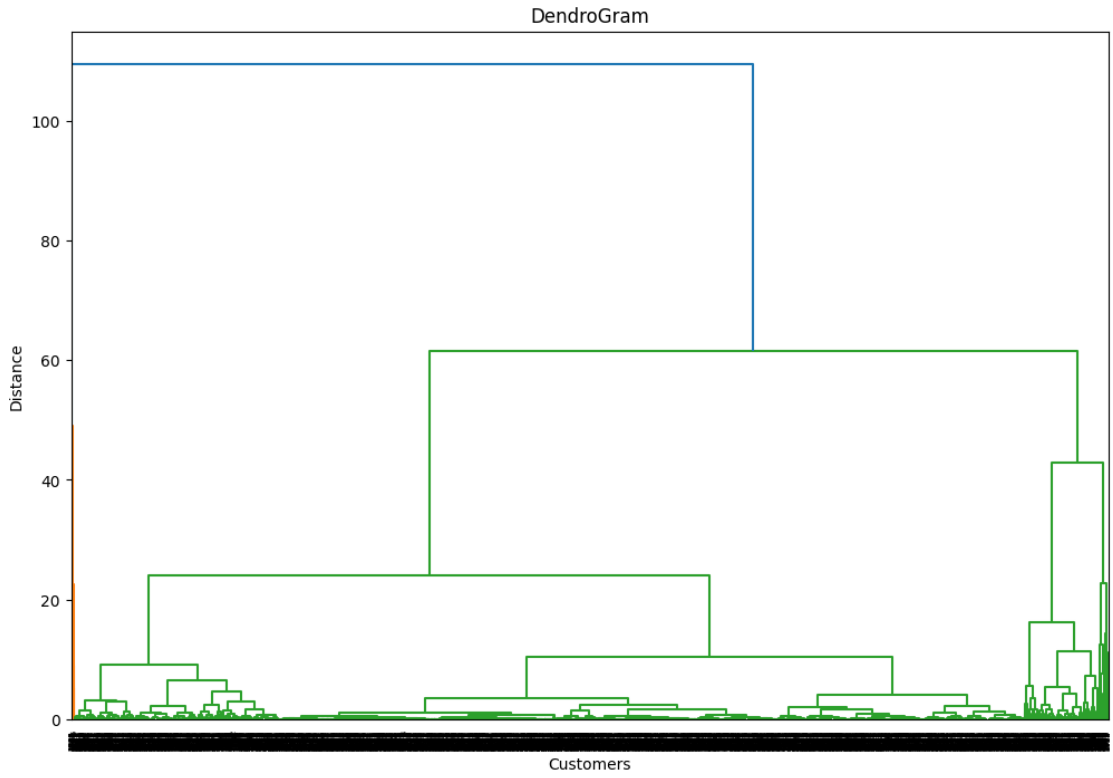
[166]:
```
              0         1
0    -0.453507 -0.117699
1     0.457734 -0.027391
2     0.094421 -0.185711
3    -0.293936 -0.334746
4    -0.456583 -0.239712
…         …         …
4367 -0.487938 -0.221632
4368 -0.494173 -0.217899
4369 -0.373788 -0.040151
4370  0.637790  0.982768
4371 -0.055324 -0.223991

[4372 rows x 2 columns]
```

[167]:
```
print(pca.explained_variance_ratio_)
```

```
[0.79651073 0.17739123]
```

[168]:
```
plt.figure(figsize=(12,8))
linked = linkage(final_data,method='ward')
dendrogram(linked)
plt.title("DendroGram")
plt.xlabel("Customers")
plt.ylabel("Distance")
plt.show()
```




[169]:
```
hc = AgglomerativeClustering(
    n_clusters = 3,
    metric = 'euclidean',
    linkage = 'ward'
)
labels = hc.fit_predict(final_data)
labels
```

[169]: array([2, 2, 2, …, 2, 1, 2], shape=(4372,))

[170]:
```
customer_data['Cluster'] = labels
customer_data
```

[170]:

| CustomerID | TotalQuantity | TotalSpent | # of Invoices | Cluster |
|---|---|---|---|---|
| 12346.0 | 0 | 0.00 | 2 | 2 |
| 12347.0 | 2458 | 4310.00 | 7 | 2 |
| 12348.0 | 2341 | 1797.24 | 4 | 2 |
| 12349.0 | 631 | 1757.55 | 1 | 2 |
| 12350.0 | 197 | 334.40 | 1 | 2 |
| … | … | … | … | … |
| 18280.0 | 45 | 180.60 | 1 | 2 |
| 18281.0 | 54 | 80.82 | 1 | 2 |
| 18282.0 | 98 | 176.60 | 3 | 2 |
| 18283.0 | 1397 | 2094.88 | 16 | 1 |
| 18287.0 | 1586 | 1837.28 | 3 | 2 |

[4372 rows x 4 columns]

[171]:
```
customer_data['Cluster'].value_counts()
```

[171]:
```
Cluster
2    4007
1     353
0      12
Name: count, dtype: int64
```

[172]:
```
print("Customers with Good Spending: \n\n")
customer_data[customer_data['Cluster'] == 1]
```

Customers with Good Spending:

[172]:

| CustomerID | TotalQuantity | TotalSpent | # of Invoices | Cluster |
|---|---|---|---|---|
| 12395.0 | 2113 | 2998.28 | 15 | 1 |
| 12409.0 | 5543 | 11056.93 | 7 | 1 |
| 12428.0 | 3503 | 7877.20 | 12 | 1 |
| 12431.0 | 2868 | 6416.39 | 18 | 1 |
| 12433.0 | 11071 | 13375.87 | 7 | 1 |
| … | … | … | … | … |
| 18226.0 | 3459 | 5192.10 | 14 | 1 |
| 18229.0 | 2488 | 7276.90 | 20 | 1 |
| 18241.0 | 1047 | 2058.09 | 18 | 1 |
| 18251.0 | 7824 | 4314.72 | 1 | 1 |
| 18283.0 | 1397 | 2094.88 | 16 | 1 |

[353 rows x 4 columns]

[173]:
```
print("Customers with Normal Spending: \n\n")
customer_data[customer_data['Cluster'] == 2]
```

```
Customers with Normal Spending:
```

[173]:

| CustomerID | TotalQuantity | TotalSpent | # of Invoices | Cluster |
|---|---|---|---|---|
| 12346.0 | 0 | 0.00 | 2 | 2 |
| 12347.0 | 2458 | 4310.00 | 7 | 2 |
| 12348.0 | 2341 | 1797.24 | 4 | 2 |
| 12349.0 | 631 | 1757.55 | 1 | 2 |
| 12350.0 | 197 | 334.40 | 1 | 2 |
| … | … | … | … | … |
| 18278.0 | 66 | 173.90 | 1 | 2 |
| 18280.0 | 45 | 180.60 | 1 | 2 |
| 18281.0 | 54 | 80.82 | 1 | 2 |
| 18282.0 | 98 | 176.60 | 3 | 2 |
| 18287.0 | 1586 | 1837.28 | 3 | 2 |

[4007 rows x 4 columns]

[174]:
```
print("Customers with Highest Spending: \n\n")
customer_data[customer_data['Cluster'] == 0]
```

```
Customers with Highest Spending:
```

[174]:

| CustomerID | TotalQuantity | TotalSpent | # of Invoices | Cluster |
|---|---|---|---|---|
| 12415.0 | 77242 | 123725.45 | 26 | 0 |
| 12748.0 | 24210 | 29072.10 | 224 | 0 |
| 13694.0 | 61803 | 62653.10 | 60 | 0 |
| 14156.0 | 57025 | 113384.14 | 66 | 0 |
| 14298.0 | 58021 | 50862.44 | 45 | 0 |
| 14646.0 | 196719 | 279489.02 | 77 | 0 |
| 14911.0 | 77180 | 132572.62 | 248 | 0 |
| 16684.0 | 49390 | 65892.08 | 31 | 0 |
| 17450.0 | 69029 | 187482.17 | 55 | 0 |
| 17511.0 | 63012 | 88125.38 | 46 | 0 |
| 17841.0 | 22865 | 40340.78 | 169 | 0 |
| 18102.0 | 64122 | 256438.49 | 62 | 0 |

[177]:
```
plt.figure(figsize=(12,7))

plt.scatter(
```

```
    final_data[labels==0,0],
    final_data[labels==0,1],
    label='Highest Spending'
)

plt.scatter(
    final_data[labels==1,0],
    final_data[labels==1,1],
    label='Good Spending'
)

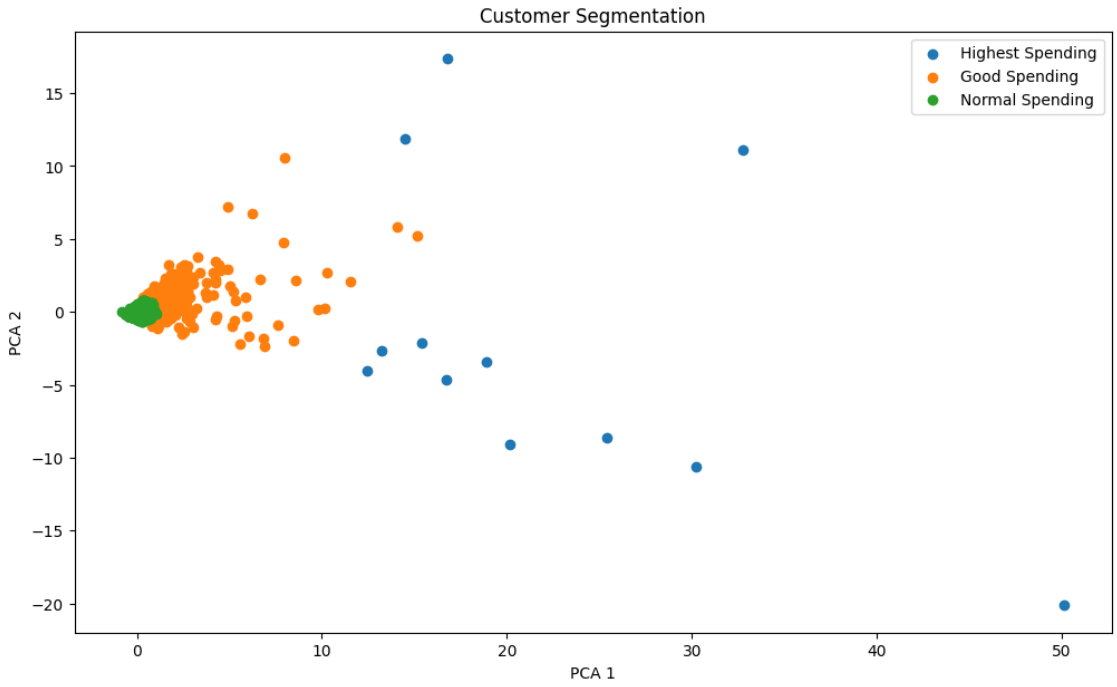
plt.scatter(
    final_data[labels==2,0],
    final_data[labels==2,1],
    label='Normal Spending'
)

plt.xlabel("PCA 1")
plt.ylabel("PCA 2")

plt.title("Customer Segmentation")

plt.legend()

plt.show()
```



[ ]: